STIVEN CATALYST

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# MANAGEMENT REVIEW

Management without theatre.




STRATEGY

## THE BOTTLENECK: THEORY OF CONSTRAINTS

Herbie on the scout hike, what published TOC cases reported, the five focusing steps, what 42 healthcare projects found, three measures instead of efficiency, and a card for your own bottleneck.

**69%** was the average cut in lead time in 32 published applications of the theory of constraints. None of the reports the authors found described a failure.

BALDERSTONE & MABIN, 1998

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NO. 54 · STRATEGY

# In this issue

Every operation has one step that sets the pace for all the others. Improve anything else and the output stays the same. The work is to find that step, use it to the full, and notice when it moves.

In 1984 Eliyahu Goldratt and Jeff Cox told the theory of constraints as a novel, The Goal. Published applications reported average cuts in lead time of around 70%, though not one report of a failure turned up. The five focusing steps show the order of the work. In 42 healthcare projects, most reported gains, and in one the queue moved to the next ward. Three measures replace local efficiency.

*Stiven Janaqi* EDITOR · STIVEN CATALYST


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## HOW TO READ THIS ISSUE

**Figure**
Every figure has its source and year at the foot of its page.

**Our reading**
Where the editor interprets rather than the research, it says so.

**Practice**
The steps and the card are proposals to try, not research results.

COVER STORY

# Herbie sets the pace

In The Goal (1984), plant manager Alex Rogo leads his son's scout troop on a hike. The line stretches out. The boys cannot overtake the one in front, and the slowest of them, Herbie, decides how fast the whole troop moves.

**01**

**Dependent events**

each hiker can go only as fast as the one ahead allows

**02**

**Fluctuations**

the delays add up along the line; the gains do not make up for them

**03**

**Bottleneck**

in Goldratt's terms, any resource with less capacity than the demand on it

Alex puts Herbie at the front, and the boys share out his heavy pack. The troop stays together and walks faster. Goldratt wrote the novel for workers in plants using his scheduling software, OPT. Its schedules left some stations idle; measured on their own efficiency, workers kept busy and built stock.

OUR READING

A busy team and a moving line are two different things. The line goes at Herbie's pace, however fast the others walk.

Sources: Eliyahu M. Goldratt & Jeff Cox, North River Press, 1984 (via Watson, Blackstone & Gardiner, 2007; LitCharts); Kevin J. Watson, John H. Blackstone & Stanley C. Gardiner, Journal of Operations Management 25(2), 2007

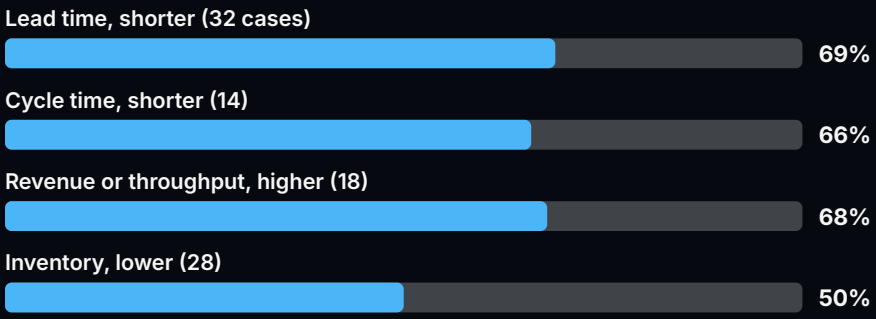
The hike is fiction from the novel; we know it from summaries and did not see the book itself. Why the book was written follows Watson, Blackstone and Gardiner (2007).

THE NUMBERS

# What the cases reported

Steven Balderstone and Victoria Mabin, at Victoria University of Wellington, searched the literature on the theory of constraints. Of over 100 case reports, 77 companies gave figures. The averages, with the number of cases behind each:

MEAN CHANGE REPORTED IN PUBLISHED TOC APPLICATIONS, 1998




In their 2003 article, with over 80 applications, the means were close: lead time 70% shorter, cycle time 65% shorter, inventory 49% lower. In both searches the authors found no report of a failure.

OUR READING

Published successes are a selected sample: they show what is possible. What to expect from an attempt of your own is another question.

Sources: Steven J. Balderstone & Victoria J. Mabin, Victoria University of Wellington, conference paper, 1998; Victoria J. Mabin & Steven J. Balderstone, International Journal of Operations & Production Management 23(6), 2003 (via Panizzolo, 2016); Roberto Panizzolo, International Journal of Industrial Engineering and Management 7(1), 2016

Reports by the companies or their authors, mostly from manufacturing in North America; the revenue mean leaves out one rise of 600%. The 2003 means as a later study cites them.

THE MODEL

# Five focusing steps

The Goal sets out the order of the steps. Later accounts add two first: define the system and its purpose, and choose measures tied to that purpose.

1. **Identify the constraint**
   the one step that limits the output of the whole
2. **Exploit it**
   get the most from it with the resources you already have
3. **Subordinate everything else**
   the other steps work at its pace, not at their own
4. **Elevate it**
   add capacity there, only if more output is still needed
5. **Do not let inertia set in**
   if the constraint has moved, start again

A constraint can be physical, when capacity is below demand; the market, when demand is below capacity; or a policy, a written or unwritten rule that holds the system back.

OUR READING

Exploit comes before elevate. The cheapest hour of capacity is the one the bottleneck loses today, waiting.

Sources: Eliyahu M. Goldratt & Jeff Cox, North River Press, 1984 (via Watson, Blackstone & Gardiner, 2007; LitCharts); Theory of Constraints Institute, 2021; Kevin J. Watson, John H. Blackstone & Stanley C. Gardiner, Journal of Operations Management 25(2), 2007

Steps after the Theory of Constraints Institute; the prerequisites and the three kinds of constraint after Watson et al. (2007), who cite the APICS Dictionary. Interpretation: the editors.

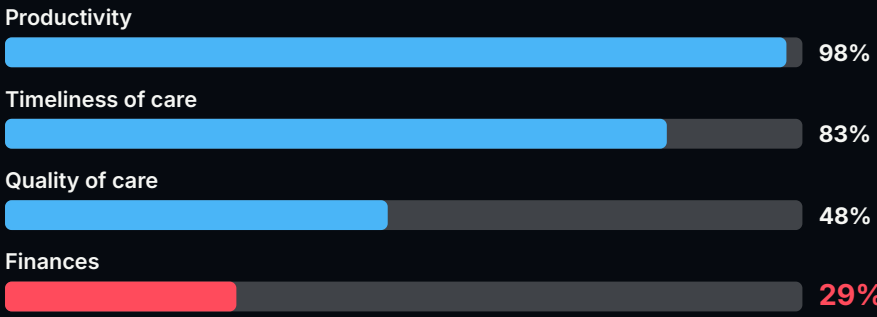
More in the essay High-volume days: the standard under pressure

WHAT THE RESEARCH SAYS

# When the queue moves on

Gabriel Bacelar-Silva, James Cox and Pedro Rodrigues reviewed 42 uses of the theory of constraints in healthcare, from journals, theses and conference talks, up to 2020. The five focusing steps were the method used most, in 76%. Share of projects that reported a gain:

42 TOC PROJECTS IN HEALTHCARE, 2020




Where they were measured, waiting times fell by half on average. In three, one measure did not change. In one, more patients got through, and the ward and the follow-up care after it became congested.

OUR READING

Lift one bottleneck and the queue moves to the next. That is why the fifth step exists.

Source: Gabriel M. Bacelar-Silva, James F. Cox III & Pedro P. Rodrigues, Health Systems 11(1), 2020

The authors call their review exploratory: few projects, data often incomplete, no statistical tests, and most applied only parts of the method.

HOW IT IS MEASURED

# Three measures, one pace

In place of the efficiency of each station, The Goal measures the plant as a whole with three numbers. Its aim: raise throughput while cutting inventory and operating expense.

**01**

**Throughput**

the rate at which the system makes money through sales

**02**

**Inventory**

the money tied up in things it intends to sell

**03**

**Operating expense**

the money spent to turn inventory into throughput

Keeping to the schedule is a fair measure at the constraint. Elsewhere, running at full output only builds stock.

HYPOTHETICAL EXAMPLE, AN EVENING SHIFT IN A DISPATCH AREA

| | |
|---|---|
| **Constraint** | the loading ramp: 4 doors, 16 door-hours |
| **Lost there** | 2.5 door-hours waiting for trolleys |
| **Elsewhere** | picking at 105% of plan, 30 trolleys queuing |

Picking looks best and the ramp loses the most. Shift and figures are invented.

Sources: Eliyahu M. Goldratt & Jeff Cox, North River Press, 1984 (via Watson, Blackstone & Gardiner, 2007; LitCharts); Kevin J. Watson, John H. Blackstone & Stanley C. Gardiner, Journal of Operations Management 25(2), 2007

The three measures and their definitions are Goldratt's, as Watson et al. (2007) and a textbook glossary give them. The example is the editors'.

TOOL OF THE ISSUE

# The bottleneck card

One process, one constraint. Look for where the work waits rather than where people look busiest, and write down the date when you will check whether it has moved.

**01** **Process and goal** where it starts and ends; what output counts

**02** **1. The constraint** where work queues; physical, market or a rule

**03** **2. Hours it loses** waiting, breaks, rework; what changes at no cost

**04** **3. What the others change** release, pace and priorities set by the constraint

**05** **4. Added capacity** only if still short after step 2; what it costs

**06** **5. Check again** date; where the queue is now

Sources: Theory of Constraints Institute, 2021; Gabriel M. Bacelar-Silva, James F. Cox III & Pedro P. Rodrigues, Health Systems 11(1), 2020

Proposed by the editors, on the five focusing steps and the findings above.

Open the tool: Delay Analyzer stivencatalyst.com/tools/delay-analyzer/

SOURCES AND METHOD

# Every figure has a source.

The figures in this issue come from the sources below. The year shows how recent each one is.


**The Goal: A Process of Ongoing Improvement**
Eliyahu M. Goldratt & Jeff Cox, North River Press, 1984 · via Watson, Blackstone & Gardiner, 2007; LitCharts

**The evolution of a management philosophy: The theory of constraints**
Kevin J. Watson, John H. Blackstone & Stanley C. Gardiner, Journal of Operations Management 25(2), 2007
https://doi.org/10.1016/j.jom.2006.04.004

**A Review of Goldratt's Theory of Constraints (TOC): lessons from the international literature**
Steven J. Balderstone & Victoria J. Mabin, Victoria University of Wellington, conference paper, 1998
https://www.tocinstitute.org/uploads/1/2/7/9/12796657/toc_impact_study.pdf

**The performance of the theory of constraints methodology: Analysis and discussion of successful TOC applications**
Victoria J. Mabin & Steven J. Balderstone, International Journal of Operations & Production Management 23(6), 2003 · via Panizzolo, 2016
https://doi.org/10.1108/01443570310476636

**Theory of Constraints (TOC) Production and Manufacturing Performance**
Roberto Panizzolo, International Journal of Industrial Engineering and Management 7(1), 2016
https://www.research.unipd.it/retrieve/e14fb26b-f0bc-3de1-e053-1705fe0ac030/ijiem_vol7_no1_3.pdf

**The Five Focusing Steps (POOGI)**
Theory of Constraints Institute, 2021
https://www.tocinstitute.org/five-focusing-steps.html

**Outcomes of managing healthcare services using the Theory of Constraints: A systematic review**
Gabriel M. Bacelar-Silva, James F. Cox III & Pedro P. Rodrigues, Health Systems 11(1), 2020
https://doi.org/10.1080/20476965.2020.1813056


## EDITORIAL METHOD

Each figure was checked for its year, its publisher and what it measures, in the full text where it could be opened. A source read only in a summary, an abstract or excerpts is marked “via”. A figure not confirmed in what was read is marked so on its page, or left out. The editor's interpretation is marked “Our reading”.



# MANAGEMENT REVIEW

Management without theatre.

Every issue, one management question, checked against the best research.

## ALL ISSUES

stivencatalyst.com/magazine/management-review.html

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Edited by Stiven Janaqi, who answers for it; “the editors” in this publication means him. Its texts, translations, layout and the checking of figures against their sources are done in working sessions with Claude, an AI model by Anthropic.
stivencatalyst.com/how-this-is-made.html